LIVER DISORDERS - A STATISTICAL STUDY

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IN

STATISTICS

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1 INTRODUCTION:

Liver, a vital organ of the the human body, is responsible for various important functions such as metabolism, protein synthesis, decomposition of red blood cells, detoxification, glycogen storage, and production of biochemical and hormone.

Excessive alcohol consumption, obesity, autoimmune diseases, and genetic predispositions, exposure to viral infections like hepatitis B and C can cause different forms of known liver disease such as Fatty Liver disease (FAD), Liver Cirrhosis, Liver failure, Hepatitis, etc . The prevalence of liver disorders varies geographically, with certain regions experiencing higher chances of being affected due to factors such as endemic viral hepatitis or dietary habits. In India, liver disorders are a significant public health concern, with conditions like hepatitis B and fatty liver disease being particularly prevalent.The most dreadful among them all is the chance of being affected by Liver cancer. Liver cancer is not easily identified and can progress silently without causing noticeable symptoms . The liver is able to maintain regular activity even if there are some damage and if there are a few irregularities in the functioning of the liver, the bring about non-specific or overlapping symptoms such as fatigue,abdominal discomfort, nausea, jaundice (yellowing of the skin and eyes), and unexplained weight loss which are generally observed is various other diseases too. Hence detection and early diagnosis of these diseases becomes the most important step.

Bilirubin which is a major factor in deciding the liver functioning is our main concern and also the influence of other factors such as Alkaline Phosphate, Albumin ,Total protein, Sgpt , Sgot , A/G Ratio on Direct Bilirubin.

2 Data Description and Data Source:

The considered data set he dataset comprising 583 patient records collected from the NorthEast of Andhra Pradesh, India.The current report considers a real data set of 584 subjects with 9 continuous variables and 2 attribute characters. The patients are divided into two groups liver patients (416) and non-liver patients (167). The data set contains 441 male and 142 female patients. Source of the data set is the UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/225/ilpd+indian+liver+patient+dataset

2.1 Data Variables:

There are 11 variables per patient: Age, Gender, Total Bilirubin, Direct Bilirubin, Total proteins, Albumin, ALB,A/G ratio, SGPT, SGOT and Alkphos.

2.1.1 Description of the variables:

1. **Dir_Bilirubin:** Direct bilirubin (sometimes referred to as conjugated) is the form of bilirubin which has been conjugated with glucoronic acid and is excreted in the bile. Measurement of this metabolite is of assistance in diagnosis and monitoring of the many disease states associated with raised bilirubin.Normal range of the Direct Bilirubin in blood test range from 0 to 0.3 mg/dL in adults.

2. **AlkPhos:** Alkaline phosphatases (Alkphos) are a group of isoenzymes located on the outer layer of the cell membrane. They catalyze the hydrolysis of organic phosphate esters found in the extracellular space.Alkaline Phosphate is found in the cytosol of liver cells and the canalicular membrane of hepatocytes.

3. **ALB:** Albumin is a protein made by our liver.It is the most abundant circulating protein found in plasma. It represents half of the total protein content (3.5 g/dL to 5 g/dL) of plasma in healthy human patients. Albumin is synthesized by liver hepatocytes and rapidly excreted into the bloodstream at the rate of about 10 gm to 15 gm per day.

4. **TP:** Serum total protein, also known as total protein, is a clinical chemistry parameter representing the concentration of protein in serum.The total protein test measures the total amount of two classes of proteins found in the fluid portion of your blood. These are albumin and globulin.The normal range is 6.0 to 8.3 grams per deciliter (g/dL) or 60 to 83 g/L.

5. **A/G ratio:** The A/G ratio, or albumin/globulin ratio, is a laboratory test that measures the relative amounts of two major groups of proteins in the blood: albumin and globulins. A normal A/G ratio is between 1.1 and 2.5. A low A/G ratio may be due to kidney disease, liver disease and further chronic infections.

6. **SGPT and SGOT:** SGPT (serum glutamic pyruvic transaminase) and SGOT (serum glutamic oxaloacetic transaminase) are enzymes found in the liver cells.Normal level of SGOT ranges from 5 to 40 units per liter of serum and normal SGPT levels - 7 to 56 units per liter of serum.

3 Literature Review

3.1 Why is Direct Bilirubin crucial for individuals afflicted with liver disorders?

Bilirubin is the major pigment formed by breakdown of hemoglobin, myoglobin, peroxidases, and cytochromes[1].Bilirubin that circulates within the body is either in conjugated or unconjugated form. Unconjugated bilirubin, also known as indirect bilirubin, is a water-insoluble compound bound to albumin. On the other hand, direct bilirubin, often referred to as conjugated bilirubin, is bilirubin that has undergone conjugation with glucuronic acid and is subsequently excreted in bile.This direct bilirubin is more crucial to the individuals prone to liver disorders [2]

This Direct Bilirubin plays a pivotal role in assessing the liver function and identifying potential issues related to the excretion of bilirubin.Liver diseases, such as hepatitis or cirrhosis, can impact the liver's ability to process and excrete bilirubin. Elevated levels of Direct Bilirubin can thus serve as an indicator of liver dysfunction. Furthermore, it can be a crucial factor in indicating Jaundice, Bile duct obstruction etc. Interpreting direct bilirubin levels alongside other liver function tests is imperative for accurate diagnosis and subsequent treatment planning.

3.2 A comparative study of Direct Bilirubin and other independent variables:

3.2.1 Direct Bilirubin and Gender:

In a normal adult, Direct Bilirubin measures less than 0.3 mg per dL of blood. Direct Bilirubin as recorded in our data was a continuous variable. To analyze this association with the Gender, we convert this into two categories ">0.3 mg" and "< 0.3 mg". we further represented this two categories i.e. the level of Direct Bilirubim ">0.3 mg" is represented by "1" and the "< 0.3 mg" as "0". Now, The two way frequency table of Direct Bilirubin and Gender(Female represented as "1" , Male as "0") was visualized as a barplot mentioned in the figure (1).

- **Interpretation from the Barplot:**

In our data, the barplot shows the gender centric presence of Direct Bilirubin from which we can clearly interpret that 69 percent of the males are mostly prone to the liver malfunctioning thus subjected to various liver disorders.

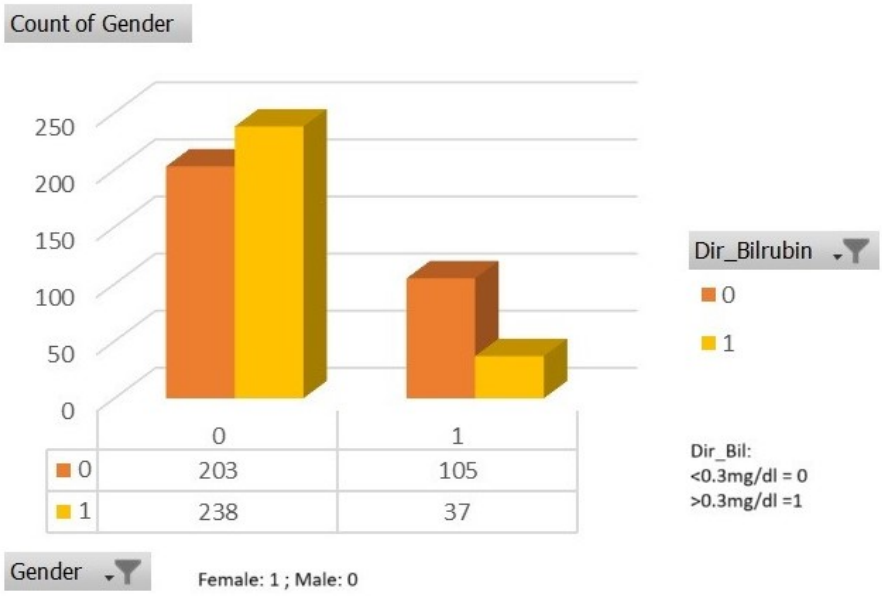

Figure 1: Direct Bilirubin v/s Gender

3.2.2 Direct Bilirubin and Alkaline Phosphate:

As mentioned earlier in the above sub-section,if the value of Direct Bilirubin in an adult exceeds 0.3 mg is represented as 1 and otherwise. Alkaline Phosphate are the group of iso-enzymes present in the outer layer of the cell membrane.The normal range of Alkaline phosphate in an adult is 44-147 IU/L.Any deviation from the normal levels of Alkaline Phosphate in blood is a an indicative to potential liver or other chronic disorders[3, 4, 5].The variable Alkaline Phosphate in our data is a continuous variable, so we categorise the data as follows: Range of Akaline Phosphate between 44-147 IU/L as "0" and > 147 IU/L as "1". The Barplot as shown in the figure (2).

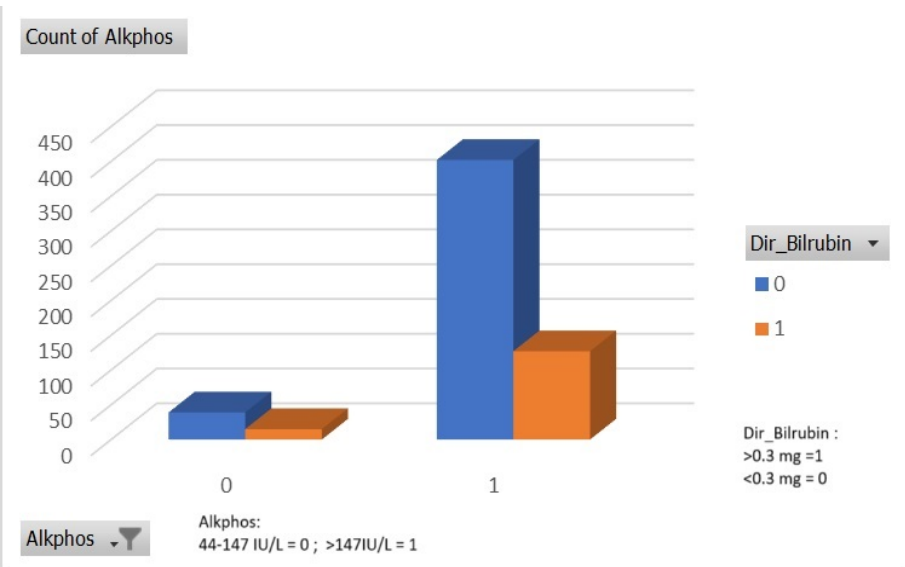

Figure 2: Direct Bilirubin v/s Alkaline Phosphate

- **Interpretation from the Barplot:**

The above bar plot thus infers that the even in the presence of high level of Alkaline Phosphate i.e > 147 IU/L (as marked by 1 in the barplot), the level of Direct Bilirubin remains in its normal range in majority of patients .Thus this pictorially shows no association between the two variables. To have a confirmation over the fact we perform Chi-squared test of Association in the later sections.

3.3 Direct Bilirubin and Albumin:

Albumin is a circulating protein in the blood plasma. The data of albumin in the blood plasma of patients recorded in the dataset is a continuous variable[6, 7] .According to the normal levels of Albumin present in an adult we categorise it into three groups and classify as follows "<3.5 g/dl" as "0" , "3.5-5.5 g/dL" as "1" and the Albumin level ">5.5 g/dL" as "2". Thus the Barplot for Direct Bilirubin and Albumin (3) is given as follows:

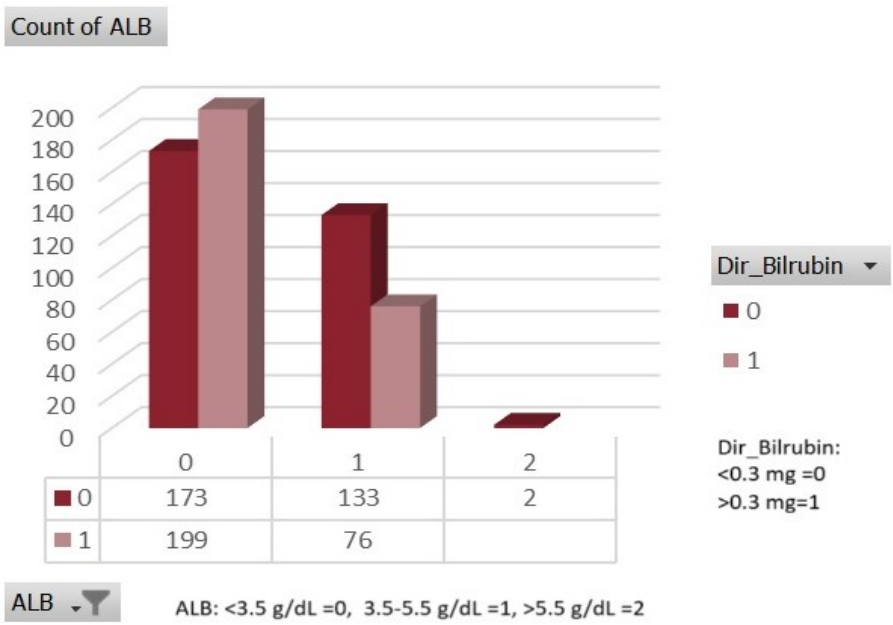

Figure 3: Direct Bilirubin v/s Albumin

- **Interpretation from the Barplot:**

From the above Barplot it is quite evident that when the Albumin levels are normal i.e. lying between 3.5-5.5 g/dl, there is no abnormality in the levels of Direct Bilirubin seen as well.Moreover, when the level of Albumin is slightly below the normal range, i.e. < 3.5 g/dl, we see a slight elevation in the level of Direct Bilirubin.This suggests a possible correlation between Albumin and Direct Bilirubin levels, thus implying that any alteration in Albumin levels may influence Direct Bilirubin levels. To have a confirmation over the fact we further perform tests to check the degree of association between the two.

3.3.1 Direct Bilirubin and Total Proteins:

Next, we investigate whether there is a significant relationship between Direct Bilirubin and Total Proteins. Total Proteins, as recorded in our dataset, represent a continuous variable indicating the concentration of two primary classes of proteins—albumin and globulin—present in the blood's serum component.An increased amount of the Direct Bilirubin and low level of Total Protein in blood can both lead to potential liver disorders [6, 7].Hence, we are interested in checking their inter association.To facilitate our analysis, we opt to discretize the Total Proteins variable into two distinct categories. Samples falling within the range of below are classified as "0" , between 6.0 to 8.3 g/dL are classified as "1", while those surpassing 8.3 g/dL are categorized as "2". Subsequently, we construct a two-way table to systematically examine any potential relationships between Direct Bilirubin and the categorized Total Proteins. The barplot (4) below shows the following result:

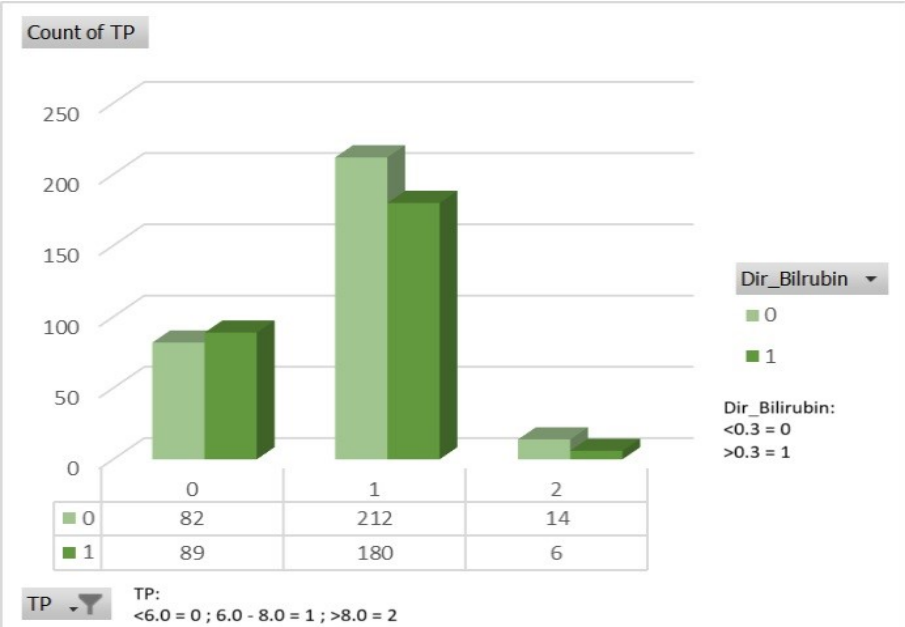

Figure 4: Direct Bilirubin v/s Total Proteins

- **Interpretation from the Barplot:**

From the above barplot, we can interpret that when the levels of Total Protein are normal i.e. lying between 6-8.3 g/dl, there is no abnormality in the levels of Direct Bilirubin seen. Moreover, when the level of Total Protein is slightly below the normal range, i.e. < 6.0 g/dl, we see that the level of Direct Bilirubin remains more or less stable.This suggests that there might not be any possible correlation between Total Protein and Direct Bilirubin levels.

3.3.2 Direct Bilirubin and SGOT and SGPT:

Expanding this, we now delve into the potential interrelation between Direct Bilirubin and Serum Glutamic Oxaloacetic Transaminase (SGOT), also known as Aspartate Aminotransferase (AST) and Serum Glutamate Pyruvate Transaminase (SGPT), also known as Alanine Aminotransferase (ALT). SGOT and SGPT are enzymes found primarily in the liver, heart etc. with its levels in the blood often serving as indicators of liver and heart health[6]. Direct Bilirubin, as we've previously

discussed, is a crucial marker of liver function.Given the clinical relevance of Direct Bilirubin, SGOT and SGPT in assessing liver function, we aim to discern any potential relationship between these variables. Considering the normal range of SGOT, we further categorize the data as the samples falling between the normal range in an adult i.e. 5-40 IU/L is labelled as "0" and those exceeding this range is marked as "1" and similarly levels of SGPT between 7-56 IU/L is denoted as "0" and subsequently those transcending this range is marked as "1".The barplot showing the correlation between Direct Bilirubin and Sgot is given by Figure (5) and the association between Direct Bilirubin and Sgpt is visualized by fig (6)

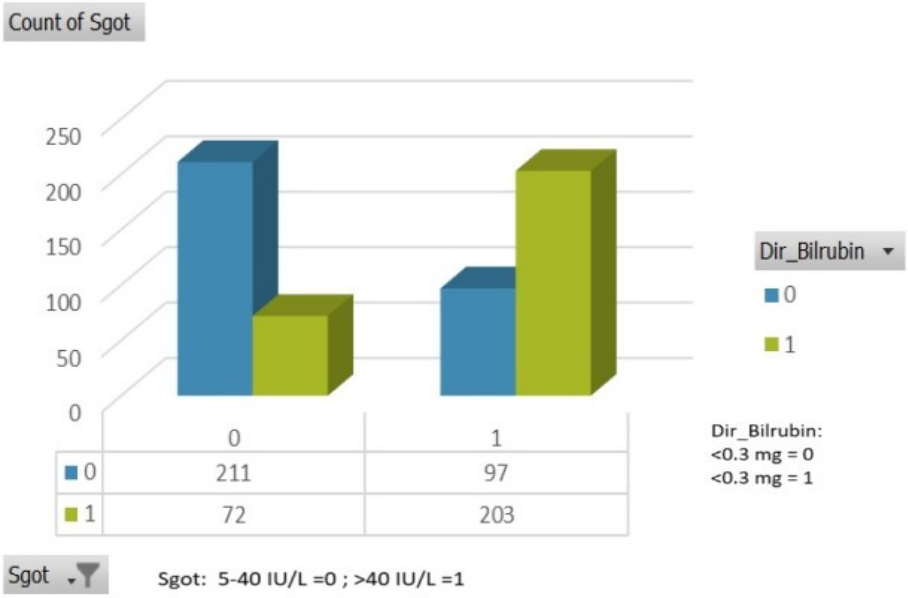

Figure 5: Direct Bilirubin v/s Sgot

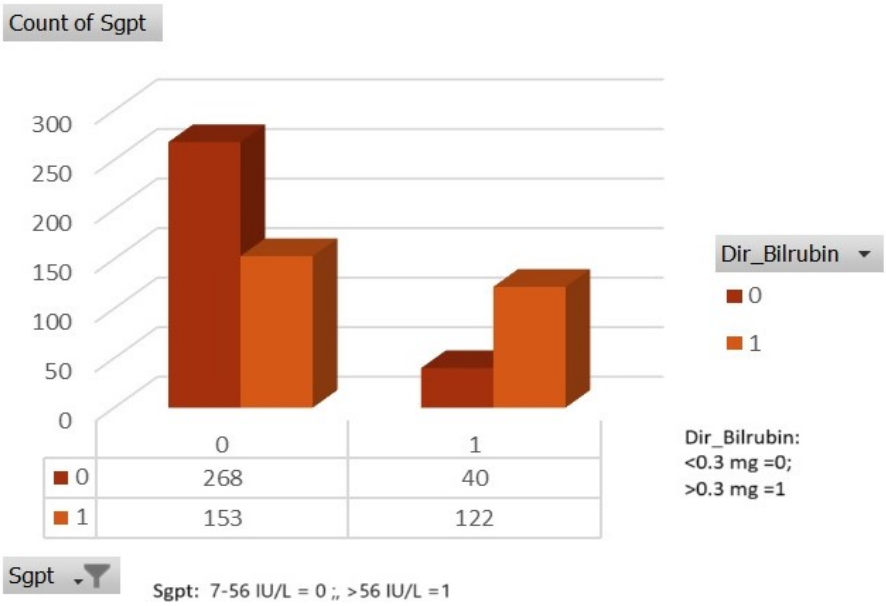

Figure 6: Direct Bilirubin v/s Sgpt

- **Interpretation from the Barplot:**

The above Barplot clearly indicates a direct correlation between Direct Bilirubin and Sgot. From the plot we can infer that when the levels of Sgot lies within the normal range, Direct Bilirubin levels also remain within normal parameters. Again, similarly a noticeable elevation in the levels of Sgot results in the increased levels of Direct Bilirubin as well. A similar pattern is observed in case of Direct Bilirubin and Sgpt as provided by the plot below.

3.3.3 Direct Bilirubin and A/G Ratio:

We further study to check the substantial interrelation between Direct Bilirubin and A/G Ratio. A/G Ratio shows the relative measure of the amount of two major protein group- albumin and globulin. An increase amount of both the Direct Bilirubin and low A/G Ratio can lead to potential liver disorders[8]. Hence, we are interested in checking its inter association. Considering the normal range of A/G Ratio in blood, we categorize the sample as in the the samples falling in the normal range i.e. 1.1-2.5 is considered as "0" and the samples below this range are marked as "1". Thus we again form a two-way table and further study its association. The barplot (7) is given below:

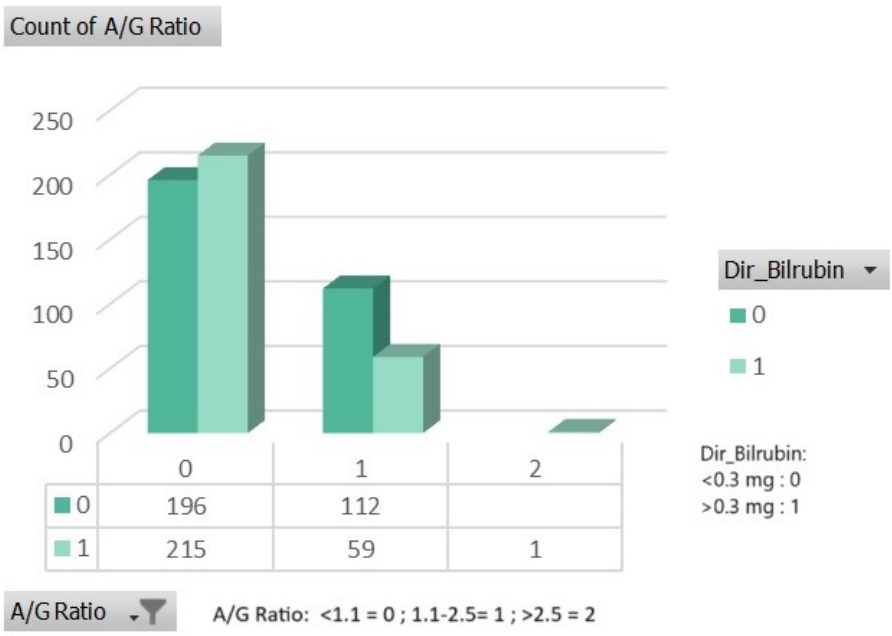

Figure 7: Direct Bilirubin v/s A/G Ratio

- **Interpretation from the Barplot:**

The bar plot above visually represents that a decrease in the Albumin and Globulin Ratio (A/G Ratio) correlates with an increase in Direct Bilirubin levels beyond the normal range thus indicating a negative correlation between the two variables.

4 Odds Ratio:

An odds ratio (OR) is a measure of association between an exposure and an outcome. The OR represents the odds that an outcome will occur given a particular exposure, compared to the odds of the outcome occurring in the absence of that exposure.
From our data we calculate the Odd and the Odds Ratio between Male and Female and perform a comparative study.Thus by this method we can provide a relative study as to which population is

more prone to be affected by Liver malfunctioning.
Through this Odds Ratio we try to portray a gender centric influence on various significant factor or variables in the data responsible mainly for any disorders in the functioning of the liver.

4.1 Test on Direct Bilirubin:

In our data, the number of Males having high levels of Direct Bilirubin is 238 whereas 37 females out of the 142 females are subjected to high values of Direct Bilirubin.The two way table (1)below represents the following:

Gender and Direct Bilirubin		
Gender	Bil >0.3 mg	Bil <0.3 mg
Male	203	238
Female	105	37

Table 1: Two way table between Direct Bilirubin and Gender

From the above table we see that the level of Direct Bilirubin above the normal range, i.e. Bil>0.3 mg is the point of exposure in our data.
As we try to find the exposure of women or females to liver disorders in our data we consider Male group as our control group. Thus the odds of having liver dysfunction (>0.3 mg) in Female = $\frac{\text{female with Bil} > 0.3 \text{ mg}}{\text{female with Bil} < 0.3 \text{ mg}} = \frac{105}{37}$;

Similarly, the odds of having liver dysfunction (>0.3 mg) in Male = $\frac{\text{male with Bil} > 0.3 \text{ mg}}{\text{male with Bil} < 0.3 \text{ mg}} = \frac{203}{238}$;

Hence, the Odds Ratio = $\frac{105}{37} / \frac{203}{238} = 3.327$

Thus from the result given by the Odds Ratio we can interpret that for female the odds of having higher Direct Bilirubin which is in turn signifies that the odds of having Liver dysfunctional issues is 3.327 times as large as the Odds of a Male having Liver disorders.

4.2 Test on Alkaline Phosphate

The influence of gender on Alkaline Phosphate is represented by the two-way pivot table as presented below. As mentioned in the above sections, the level of the Alkaline Phosphate exceeding the value of 147 IU/L is the point of exposure in our dataset.
The table (2) below shows the following results.
Similarly, we consider the Male group as the control group as we delve on the effect of Alkaline

Gender and Direct Bilirubin		
Gender	Alkphos 44-147 IU/L	Alkphos >147 IU/L
Male	39	402
Female	15	127

Table 2: Two way table between Alkaline Phosphate and Gender

Phosphate on Females.

The Odds of having liver dysfunction in Female is $= \frac{\text{female with Alkphos} > 147 \text{ IU/L}}{\text{female with Alkphos 44-147 IU/L}} = \frac{127}{15}$

Similarly, the Odds for Male being exposed to the disease is $= \frac{\text{male with Alkphos} > 147 \text{ IU/L}}{\text{male with Alkphos 44-147 IU/L}} = \frac{402}{39}$

Thus, the Odds Ratio is given by= $\frac{127}{15}/\frac{402}{39} = 0.821$

Thus from the result given by the Odds Ratio we can interpret that for female the odds of having higher Alkaline Phosphate levels which in turn signifies that the odds of having Liver dysfunctional issues is 0.821 times the Odds of a Male having Liver disorders.

4.3 Test on Albumin:

The gender-centric influence on Albumin is represented by the two-way pivot table (3) as presented below.The decrease in the level of the Albumin below the normal level is a possible reason for liver malfunctions.

Hence, we find the odds of female and male being exposed to the liver disorders and interpret the

Gender and Albumin			
Gender	Alb < 3.5g/dL	Alb 3.5-5.5 g/dL	Alb > 5.5 g/dL
Male	292	148	1
Female	80	61	1

Table 3: Two way table between Gender and Albumin

result on the basis of the female population.
The Odds of having liver dysfunction in Female is $=\frac{\text{female with Albumin} < 3.5 \text{ g/dL}}{\text{female with Albumin 3.5-5.5 g/dL}} = \frac{80}{61}$

Similarly, the Odds for Male being exposed to the disease is $=\frac{\text{male with Albumin} < 3.5 \text{ g/dL}}{\text{male with Albumin 3.5-5.5 g/dL}} = \frac{292}{148}$

Thus, the Odds Ratio is given by= $\frac{80}{61}/\frac{292}{148} = 0.665$

Thus from the result given by the Odds Ratio we can interpret that for female the odds of having lower Albumin levels which is in turn signifies that the odds of having Liver dysfunctional issues is 0.665 times as the Odds of a Male having Liver disorders.

4.4 Test on Total Proteins:

As Total Protein is one of the significant factors in deciding the functioning of the Liver, thus we try to check the influence of Gender on Albumin. This is represented by the two-way pivot table (4) as presented below.

Taking, the Male population as the control group

Gender and Total Protein			
Gender	TP <6.0g/dL	TP 6.0-8.3 g/dL	TP > 8.3 g/dL
Male	135	295	11
Female	36	97	9

Table 4: Two way table between Gender and Total Protein

The Odds of having liver dysfunction in Female is $=\frac{\text{female with Total Protein} < 6.0 \text{ g/dL}}{\text{female with Total Protein 6.0-8.3 g/dL}} = \frac{36}{97}$

Similarly, the Odds for Male being exposed to the disease is $=\frac{\text{male with Total Protein} < 6.0 \text{ g/dL}}{\text{male with Total Protein 6.0-8.3 g/dL}} = \frac{135}{295}$

Thus, the Odds Ratio is given by= $\frac{36}{97}/\frac{135}{295} = 0.81$

Thus from the result given by the Odds Ratio we can interpret that for female the odds of having lower Total Protein levels which is in turn signifies that the odds of having Liver dysfunctional issues is 0.81 times the Odds of a Male having Liver disorders.

4.5 Test on A/G Ratio:

The gender-centric influence on A/G Ratio is represented by the two-way pivot table (5) as presented below. Since we are interested to check the effect of low levels of A/G Ratio, which suggests a liver dysfunction in female group.
Thus taking the Male population as the control group, we calculate the odds as follows.

Gender and A/G Ratio			
Gender	A/G Ratio <1.1	A/G Ratio 1.1-2.5	A/G Ratio > 2.5
Male	307	133	1
Female	104	38	0

Table 5: Two way table between Gender and A/G Ratio

The Odds of having liver dysfunction in Female is $=\frac{\text{female with A/G Ratio} < 1.1}{\text{female with A/G Ratio 1.1-2.5}} = \frac{104}{38}$

Similarly, the Odds for Male being exposed to the disease is $=\frac{\text{male with with A/G Ratio} < 1.1}{\text{male with A/G Ratio 1.1-2.5}} = \frac{307}{133}$

Thus, the Odds Ratio is given by= $\frac{104}{38}/\frac{307}{133} = 1.186$

Thus from the result given by the Odds Ratio we can interpret that for Male the odds of having lower A/G Ratio levels which is in turn an indicator that the odds of having Liver dysfunctional issues is 1.186 times as large as the Odds of a female having Liver disorders.

5 Methodology

5.1 Pearson Chi-squared Test Of Association:

The **Chi-squared Test** (also chi-square or χ^2 test) is a statistical hypothesis test used in the analysis of contingency tables when the sample sizes are large.In other words it is a statistical method used to determine if there is a significant association between two categorical variables.

- **<u>Test Statistic:</u>**

The test is based on the test statistic that quantifies the association between the row and column variables in a two-way (contingency) table consisting .This test is done under the assumption or null hypothesis H_0 that there is no association between the two variables, whereas the alternative hypothesis H_a states that the association exists. The expected frequency for each cell (i,j) is calculated as:

$$E_{ij} = \frac{\text{row sum of row } i \times \text{column sum of column } j}{\text{total sample size}}$$

Now , The observed frequencies (Oij) are the actual counts observed in each cell of the contingency table. Once we find the value of expected and observed frequencies are calculated we find the chi-square test statistic using the formula:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{1}$$

The degrees of freedom for a chi-square test depend on the dimensions of the contingency table. For a contingency table with r rows and c columns, the degrees of freedom (df) is calculated as df=$(r-1) \times (c-1)$.The p-value for the chi-square testis p(χ^2> X2), the probability of observing a value at least as extreme as the test statistic for a chi-square distribution with (r-1)(c-1) degrees of freedom.*If the value of the test statistic or the p-value is greater than the level of significance(say 0.05), then we fail to reject the null-hypothesis* H_0.

5.2 Cramer's V Test:

Cramer's V is a robust statistical measure derived from the chi-square test, specifically designed to quantify the strength and significance of the association between two categorical variables. Unlike measures such as Pearson's correlation coefficient, which is suited for continuous data, Cramer's V adapts the chi-square statistic to a 0-1 scale, offering a clear, interpretable measure of association for categorical data.

- **<u>Test Statistic:</u>**

The test is based on the test statistic that gives the measure(between 0-1) of the association. On performing the Chi-square test by using formula (1) as we mentioned in the previous section, we determine whether there a significant association between the two categorical variables. Once you have the chi-squared statistic (χ^2), you can calculate Cramer's V using the following formula:

$$\sqrt{\frac{\chi^2/n}{\min(c-1, r-1)}} \tag{2}$$

where r= no. of rows ; c= no.of columns; and n=the total no. of observations.

5.3 Chow Test:

The Chow Test is a statistical test developed by to test whether the coefficients in two different regression models on different datasets are equal.If the test determines that the coefficients are not equal between the regression lines, this means there is significant evidence that a structural break exists in the data.
We can perform a Chow test by the following steps:

- **Step1: To define the null and the alternative hypotheses:**

Suppose we fit the following regression model to our entire dataset:

$$y_t = a + bx_{1t} + cx_{t2} + \epsilon$$

Then suppose we split our data into two groups based on some structural break point and fit the following regression models for each group:
1)$y_t = a_1 + b_1x_{1t} + c_1x_{t2} + \epsilon$
2)$y_t = a_2 + b_2x_{1t} + c_2x_{t2} + \epsilon$

We would use the following null and alternative hypotheses for the Chow Test:

Null hypothesis $(H_0) : a_1 = a_2, \quad b_1 = b_2 \quad \text{and} \quad c_1 = c_2$
Alternative hypothesis (H_A) : At least one of the comparisons in Null is not equal.

If we reject the null hypothesis, it indicates that there is significant evidence supporting the presence of a structural break point in the data. This suggests that using two regression lines could potentially provide a better fit to the data compared to using just one.

- **Step 2: To calculate the Test Statistic:**

We first define the following terms as:

1)S_T : The sum of squared residuals from the total data.

2)S_1, S_2 : The sum of squared residuals from each group.

3)N_1, N_2 : The number of observations in each group.

4)k : The number of parameters.

Then we can say that the Chow test statistic is:

$$\frac{[S_T - (S_1 + S_2)]/k}{[(S_1 + S_2)/(N_1 + N_2 - 2k)]}$$

This test statistic follows the F-distribution with k and and $N_1 + N_2 - 2k$ degrees of freedom.

- **Step 3: Acceptance or Rejection of the null hypotheses:**

If the p-value associated with this test statistic is less than a certain significance level(say 0.05), we can reject the null hypothesis and conclude that there is a structural break point in the data.

6 Test Between Direct Bilirubin and other independent variable:

Now, we venture to test the association of Direct Bilirubin, taking it as the dependent variable with other factors namely, Alkaline Phosphate, Albumin, Total Proteins, Sgot, Sgpt, A/G ratio in our dataset as the independent variable:

6.1 Test Between Direct Bilirubin and Gender

The two way table for Gender and Direct Bilirubin is given as:

Direct Bilirubin and Gender		
Gender	Bil >0.3 mg	Bil <0.3 mg
Male	203	238
Female	105	37

Table 6: Two way table between Direct Bilirubin and Gender

Pearson Chi-squared Test:

To have a confirmation over the association between Direct Bilirubin and Gender, we perform the Pearson Chi-square Test. performing Chi-square test in R gives the following result as output:

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  data
X-squared = 32.47, df = 1, p-value = 1.21e-08
```

The Pearson Chi-square test in R yield the p-value 1.21e-08 which is much less than 0.05. Thus we accept the alternative hypothesis which indicates that Direct Bilirubin is dependent over Gender.

Cramer's V Test:

Now to check the measure of association between these two variables we perform the Cramer's V Statistic, given by the test statistic given by equation (2)
Now applying the Cramer's V Statistic in R, we get the calculated value as 0.24. This value thus indicates a moderate relationship between the two variables; Direct Bilirubin and Gender.

6.2 Test Between Direct Bilirubin and Alkaline Phosphate:

The two way table for Direct Bilirubin and Alkaline Phosphate is is given as follows:

Direct Bilirubin and Alkphos		
Alkaline Phosphate	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
44-147 IU/L	39	15
>147 IU/L	402	127

Table 7: Two way table between Direct Bilirubin and Alkaline Phosphate

Pearson Chi-squared Test:
To have a confirmation over the association between Direct Bilirubin and Alkaline Phosphate,we perform the Pearson Chi-square Test.The test yields the following results:

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  data1
X-squared = 0.20109, df = 1, p-value = 0.6538
```

We see that the Chi-square test in R yields the p-value as 0.6538 which is greater than the level of significance i.e. 0.05. So, we accept the null-hypothesis i.e. there is no relationship between the two variables Direct Bilirubin and Alkaline Phosphate.
Now, we further look into the association between other independent variables and Direct Bilirubin.

6.3 Test Between Direct Bilirubin and Albumin:

We further check the association between Direct Bilirubin and Albumin, if there's any. For which we need to first make a two way table for these categorical variables:

Direct Bilirubin and Albumin		
Albumin	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
<3.5g/dL L	173	199
3.5-5.5 g/dL	133	76
> 5.5 g/dL	2	0

Table 8: Two way table between Direct Bilirubin and Albumin

Pearson Chi-squared Test:
We perform the Chi-square test in R to check if there is any possible association between these two categorical variables. This test in R yields the following result as output:

```
        Pearson's Chi-squared test'

data:  data3
X-squared = 17.551, df = 2, p-value = 0.0001545
```

We clearly see that the Chi-square test in R gives the p value as 0.0001545 which is significantly less than 0.05 which leads to the decision that we accept the Alternative hypothesis implying that there is a possible association between the two variables.

Now, similarly to check the measure of association we perform the Cramer's V Test. The Cramer's V test in R gives the value of the test statistic as 0.1735 . This implies that there a small association between the two categorical variables Albumin and Direct Bilirubin.

6.4 Test Between Direct Bilirubin and Total Proteins:

We check if there is indeed any association between Direct Bilirubin and Total Proteins.
For which we need to first make a two way table for these categorical variables:

Direct Bilirubin and Total proteins		
Albumin	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
<6.0g/dL L	82	89
6.0-8.3 g/dL	212	180
> 8.3 g/dL	14	6

Table 9: Two way table between Direct Bilirubin and Total Protein

Pearson Chi-squared Test:
We perform the Chi-square test in R to check if there is any possible association between these two categorical variables. This test in R yields the following result as output:

```
    Pearson's Chi-squared test'

data:  data2
X-squared = 4.2445, df = 2, p-value = 0.1198
```

From the above result we clearly see that the p-value is greater than the level of significance 0.05 , say, thus leading to the acceptance of the null hypothesis : There is no association between the two categorical variables.

6.5 Test Between Direct Bilirubin and Sgpt:

We check if there is indeed any association between Direct Bilirubin and Sgpt. For which we need to first make a two way table for these categorical variables:

Direct Bilirubin and SGPT		
Sgpt	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
7-56 IU/L L	268	153
> 56 IU/dL	40	122

Table 10: Two way table between Direct Bilirubin and Sgpt

Pearson Chi-squared Test:

We perform the Chi-square test in R to check if there is any possible association between these two categorical variables. This test in R yields the following result as output:

```
     Pearson's Chi-squared test with Yates' continuity correction

data:  data4
X-squared = 69.725, df = 1, p-value < 2.2e-16
```

From the above result we clearly see that the p-value is much lesser than the level of significance thus leading to the acceptance of the Alternative hypothesis : There is an association between the two categorical variables.We further perform the Cramer's V Test to check the degree of association between the two variables

Cramer's V Test:

As we already verified that these two categorical variables as mentioned above are dependent or associated, so we further test on to measure the amount of association using Cramer's V Test in R. We get the calculated value of this test statistic as 0.3497. This implies that there is a moderately strong relationship between the variables

6.6 Test Between Direct Bilirubin and Sgot:

We check if there is indeed any association between Direct Bilirubin and Sgot. For which we need to first make a two way table for these categorical variables:

Direct Bilirubin and SGOT		
Sgot	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
5-40 IU/L L	211	72
> 40 IU/dL	97	203

Table 11: Two way table between Direct Bilirubin and Sgot

Pearson Chi-squared Test:

We perform the Chi-square test in R to check if there is any possible association between these two categorical variables. This test in R yields the following result as output:

```
        Pearson's Chi-squared test with Yates' continuity correction

data:  data5
X-squared = 102.5, df = 1, p-value < 2.2e-16
```

Here, we clearly see that the p-value which is given as $< 2.26e\text{-}16$ is much lesser than the level of significance i.e. 0.05(say). Thus we accept the Alternative hypothesis which implies that there is a significant association between the two categorical variables.

Cramer's V Test:

As we already verified that these two categorical variables as mentioned above are dependent or associated, so we further test on to measure the amount of association using Cramer's V Test in R. We get the calculated value of this test statistic as 0.4227. This implies that there is a moderately strong relationship between the variables.

6.7 Test Between Direct Bilirubin and A/G Ratio:

Direct Bilirubin and A/G Ratio		
Albumin	Dir_Bil >0.3 mg	Dir_Bil <0.3 mg
< 1.1	196	215
1.1-2.5	112	59
>2.5	0.5	1.5

Table 12: Two way table between Direct Bilirubin and A/G Ratio

Pearson Chi-squared Test:

We perform the Chi-square test in R to check if there is any possible association between these two categorical variables. This test in R yields the following result as output:

```
        Pearson's Chi-squared test'

data:  data6
X-squared = 16.49, df = 2, p-value = 0.0002625
```

Cramer's V Test:

As we already verified that these two categorical variables as mentioned above are dependent or associated, so we further test on to measure the amount of association using Cramer's V Test in R. We get the calculated value of this test statistic as 0.1682 with degrees of freedom 2. This implies that there is a small relationship between the variables.

7 Regression Analysis:

We already see the degree of associations between the Direct Bilirubin and the other factors through various tests in the earlier sections. As already mentioned that, Direct bilirubin, a key component in evaluating liver health, serves as a crucial biomarker in clinical assessments. However, analyzing its interactions with other independent variables (i.e. Alkaline Phosphate, Albumin, Total Protein, Sgpt etc.) is essential for comprehensive diagnostic and prognostic evaluations.
This study endeavors to compare direct bilirubin with other independent variables, exploring their relationships and potential implications. By performing a regression analysis considering Direct

Bilirubin as the dependent variable and examining these factors collectively, we aim to enhance our understanding of liver function assessment.

7.1 Multiple Linear Regression:

Here we consider a Linear Regression Model to perform the Regression Analysis. We perform a step-wise regression analysis systematically incorporating the various factors one by one. This helps us to check the level significance of each factor for each model. Furthermore, the Adjusted R^2 acts an indicator of the model's goodness of fit. Hence the model with the maximum value of the Adjusted R^2 indicates the bet fit for our data.

The Step-wise Regression table for our data is given below:

Regression Table			
Dependent Variable	**Regressors**	**Significance**	**Adjusted R^2**
Direct Bilirubin	Gender	Poorly Significant	0.008117
Direct Bilirubin	Gender Alkphos	Moderately Significant Strongly Significant	0.06277
Direct Bilirubin	Gender Alkphos Sgpt	Poorly Significant Strongly Significant Strongly Significant	0.1002
Direct Bilirubin	Gender Alkphos Sgpt Sgot	Poorly Significant Strongly Significant No Significance Poorly Significant	0.1075
Direct Bilirubin	Gender Alkphos Sgpt Sgot TP	Poorly Significant Strongly Significant No Significance Poorly Significant No Significance	0.1064
Direct Bilirubin	Gender Alkphos Sgpt Sgot TP ALB	Poorly Significant Strongly Significant Poorly Significant No Significance Stronglyb Significant Strongly Significant	0.2054
Direct Bilirubin	Gender Alkphos Sgpt Sgot TP ALB A/G Ratio	No Significance Strongly Significant Moderately Significant No Significance Strongly Significant Strongly Significant Strongly Significant	0.2395

Table 13: Step-wise Regression Table

From the Step-wise Regression table, we see that the value of Adjusted R^2 is maximum for the Linear Model given as:

$$Dir_Bilrubin \sim Gender + Alkphos + Sgpt + Sgot + TP + ALB + A_G_Ratio$$

.
On performing regression analysis using R-Programming,we get the Summary statistics of the above Linear Regression Model as follows:

```
    Call:
lm(formula = Dir_Bilrubin ~ Gender + Alkphos + Sgpt + Sgot +
    TP + ALB + A_G_Ratio, data = liver_disorder_)

Residuals:
    Min      1Q  Median      3Q     Max
-5.0968 -1.0502 -0.4268  0.1940 16.1819

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -3.0754133  0.8139314  -3.778 0.000174 ***
Gender      -0.3913820  0.2415634  -1.620 0.105741
Alkphos      0.0016904  0.0004380   3.860 0.000126 ***
Sgpt         0.0026669  0.0009257   2.881 0.004114 **
Sgot         0.0004818  0.0005898   0.817 0.414352
TP           1.8469437  0.2089006   8.841  < 2e-16 ***
ALB         -3.5144482  0.3838124  -9.157  < 2e-16 ***
A_G_Ratio    3.1264359  0.6057033   5.162 3.38e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.456 on 571 degrees of freedom
Multiple R-squared:  0.2487,        Adjusted R-squared:  0.2395
F-statistic: 27.01 on 7 and 571 DF,  p-value: < 2.2e-16
```

From the Summary Statistics derived from the multiple linear regression, we interpret that the independent factors in our dataset concerning the dependent variable Direct Bilirubin, i.e Alkphos, Sgpt, TP, ALB and A/G Ratio are significant, where Alkphos,TP, ALB and A/G Ratio shows strong significance, while that of Sgpt shows a moderate significance while the factors Gender and Sgot shows no significance.

The Adjusted R^2 value comes out to be 0.2395 for this model, which is very small, indicating a poor fit to the data. Thus we further proceed onto Generalised Regression Model.

7.2 Generalised Linear Model :

To proceed with this method of Regression, we first make the following assumptions on the variables. Firstly, we assume that the variance of Direct Bilirubin is constant within the whole range (<0.3 mg). Second, the positive data from continuous variable with constant variance is generally analysed by Log Normal or Gamma Model link function in Generalized linear models[9]. Thus, we consider these two models for our analysis.

1. **GAMMA MODEL :**

For the model,

$$Dir_Bilrubin \sim Gender + Alkphos + Sgpt + Sgot + TP + ALB + A_GRatio$$

taking the Direct Bilirubin is the dependent variable and then considering the Generalized Linear Model with Gamma as the link function in this case, we get the Summary Statistics of the regression as follows:

```
   Call:
glm(formula = Dir_Bilrubin ~ Gender + Alkphos + Sgpt + Sgot +
    TP + ALB + A_G_Ratio, family = Gamma, data = liver_disorder_)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.6290  -1.2884  -0.9888  -0.0551   4.6099

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.663e-01  2.823e-01   2.360  0.01860 *
Gender       3.089e-01  1.100e-01   2.810  0.00513 **
Alkphos     -3.198e-04  6.457e-05  -4.953 9.64e-07 ***
Sgpt        -6.117e-04  1.144e-04  -5.347 1.29e-07 ***
Sgot         1.318e-04  6.092e-05   2.163  0.03093 *
TP          -1.588e-01  2.665e-02  -5.960 4.41e-09 ***
ALB          4.469e-01  7.355e-02   6.076 2.25e-09 ***
A_G_Ratio   -8.974e-02  1.026e-01  -0.874  0.38228
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 3.493868)

    Null deviance: 1213.45  on 578  degrees of freedom
Residual deviance:  965.05  on 571  degrees of freedom
AIC: 1359.4

Number of Fisher Scoring iterations: 8
```

From the Summary Statistics derived from the above model, we interpret that the independent factors in our dataset influencing the levels of the dependent variable Direct Bilirubin, i.e Gender, Alkphos, Sgpt, Sgot, TP, and ALB are significant, where Alkphos, Sgpt, TP, ALB shows strong significance, while Gender shows a moderate significance, the factor Sgot is poorly significant whereas A/G Ratio shows no significance at all. Moreover, the AIC (Akaike Information Criterion) for this model is **1359.4**.
We further look into the regression analysis using Log-Normal Model.

1. **LOG-NORMAL MODEL :**

Considering the same model,

$$Dir_Bilrubin \sim Gender + Alkphos + Sgpt + Sgot + TP + ALB + A_GRatio$$

and further considering the Generalized Linear Model but considering "log" as the link function and family as "gaussian" in this case, we obtain the following as the summary statistic of the regression:

```
  Call:
glm(formula = log(Dir_Bilrubin) ~ Gender + Alkphos + Sgpt + Sgot +
    TP + ALB + A_G_Ratio, family = gaussian, data = liver_disorder_)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.5489  -0.7756  -0.1560   0.6508   3.6649

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.1911586  0.3690629  -5.937 5.04e-09 ***
Gender      -0.4745994  0.1095327  -4.333 1.74e-05 ***
Alkphos      0.0010255  0.0001986   5.164 3.34e-07 ***
Sgpt         0.0015071  0.0004197   3.591 0.000358 ***
Sgot         0.0002739  0.0002674   1.024 0.306261
TP           0.7239991  0.0947223   7.643 9.00e-14 ***
ALB         -1.3828584  0.1740330  -7.946 1.03e-14 ***
A_G_Ratio    0.9034735  0.2746455   3.290 0.001065 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 1.240283)

    Null deviance: 1021.7  on 578  degrees of freedom
Residual deviance:  708.2  on 571  degrees of freedom
AIC: 1777.8

Number of Fisher Scoring iterations: 2
```

From the statistics obtained from the regression as given above, we can see that most of the independent factors influencing the level of the Regrassand Direct Bilirubin as mostly significant. Gender, Alkphos, Sgpt, Sgot, TP, ALB shows a very strong significance, whereas A/G Ratio shows moderate significance while Sgot shows no significance at all.
However the AIC for this model is **1777.8** , which is comparatively larger than the value of AIC for the Gamma Model as we discussed earlier.

Hence, given the lower AIC value associated with the Gamma Model, we conclude that it offers a more optimal fit for our dataset compared to alternative model. This suggests that the Gamma Model captures the underlying pattern within our data more effectively. As a result, thus prefer to carry out any further analysis using this model.

8 Chow test:

The Chow Test is conducted to ascertain whether the regression equations differ between two qualitative attributes, with other quantitative regressors are fixed. In our study on Indian liver patients, we divide the dataset by gender, into two sub populations - Male and Female. By conducting the Chow Test on this gender-segregated data, we delve into the fact that whether there are distinct regression equations for male and female patients, even when other quantitative factors are held constant.This analysis allows us to draw conclusions specific to each gender regarding the relationship between predictor variables and liver health outcomes.
Now, from Chow test, we obtain the following results:

```
F value        d.f.1        d.f.2      P value
  1.87730056   7.00000000 565.00000000   0.07094975
```

From the results obtained above, we see that the p value is **0.07094975** which is greater than the level of significance, say 0.05. Thus, for this test we accept the null hypothesis H_0 i.e. we conclude that there is no structural break in the data and similar regression line can be fit for the both the populations Male and Female.

9 CONCLUSIONS:

From our study we infer that Direct Bilirubin which is one of the most crucial factors in determining the functioning of the liver is indeed dependent on the other independent factors i.e. Alkaline Phosphate, Albumin, Total Proteins, A/G Ratio and thus these factors do have a direct or indirect influence role in determining the levels of Direct Bilirubin and in turn plays an important role in judging the liver functions. Thus, every individual should care and have a regular check on Direct Bilirubin values at older ages since with increase in age values of total protein, Albumin and Alkaline Phosphates increases the Direct Bilirubin values.
Moreover, from our study in this project, we can also conclude that the two populations categorised in our dataset, namely, Male and Female, can be regressed through the same regression line, thus there is no structural break in the data. Also, from odds ratio , we can conclude that Male are more prone to liver dysfunctional issues compared to Female.

Data Availability

This data is taken from UC Irvine Machine Learning Repository https://archive.ics.uci.edu/.The direct link to access the data set is ILPD (Indian Liver Patient Dataset https://archive.ics.uci.edu/dataset/225/ilpd+indian+liver+patient+dataset [10]

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